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SFC Flotation Cell

LONGi MAGNET CO.,LTD.
LONGi MAGNET AUSTRALIA PTY LTD

SFC Series / Advanced Flotation Cell

■ Applications

The SFC flotation cell is designed for the recovery of fine particles (0–0.25 mm) and is widely used in:

- Coal flotation
- Graphite flotation
- Fine mineral flotation
- Difficult-to-recover mineral systems

■ Product Advantages

Two-stage flotation zone:

Lower turbulence zone ensures effective mineralization

Upper static zone improves separation accuracy

Fast flotation kinetics: The flotation kinetics rate reaches **4-5 times faster** than conventional flotation machines.

High processing capacity: Processing capacity is **2–3 times higher** than conventional flotation machines, reducing equipment footprint by **more than 50%**.

Better separation performance:

Clean coal ash reduced by 1–3%

Tailings ash increased by more than 5%

Reagent consumption reduced by over 20%

Combustible recovery increased by 3–5%



■ Operating Principle

The high-speed rotor generates strong turbulence in the mineralization zone, dispersing air into the slurry and producing fine bubbles.

Hydrophobic mineral particles attach to these bubbles and rise to the surface.

The aerated slurry then enters a static separation zone where attached particles float to form the concentrate while non-hydrophobic particles settle and are discharged as tailings.

The aerated slurry then rises into a longer static separation zone: more hydrophobic particles float into the concentrate trough, while less hydrophobic particles detach and sink. Guided by the separation baffle, inner descending slurry re-enters the separation zone, while outer slurry is discharged as tailings.



■ Technical Specifications

Types of coal	Flotation equipment	Unit processing capacity (m ³ /m ² ·h)	Coal ash content (%)	Residual coal ash content (%)	Combustible recovery rate (%)
Industrial Test Results	Traditional flotation machine	4.69	12.25	74.96	88.42
	SFC Flotation Cell	7.50	9.80	79.48	90.98
Pilot Test Results	Traditional flotation machine	5.24	13.53	67.42	89.97
	SFC Flotation Cell	15.00	11.96	73.51	91.94

Equipment selection is based on the slurry volume, and various properties of the material, such as density, slurry concentration, and viscosity, all affect the processing capacity and effectiveness. It is generally believed that reducing the processing volume within a certain range is beneficial for improving separation efficiency.

Model specifications	Effective volume (m ³)	Unit processing capability (t/m ³ ·h)	Unit slurry treatment capacity (m ³ /m ³ ·h)	Motor power (kW)	Overall Dimensions (mm)
SFC-5	5	1.5-2	15-20	90	3790X2970X4400
SFC-10	10	1.5-2	15-20	132	4390X3570X5600
SFC-20	20	1.5-2	15-20	160	5140X4320X6100

The technical parameters provided are for reference only.

Final equipment selection should be based on site conditions including slurry density, particle size distribution, slurry concentration, and viscosity.